

# Chapter 3. REGISTERS

## Last Week

### Important operations

- Registers with Parallel-Load

- Registers with Clear

- Registers with Increment

## This Week

### Combining Load and Clear inputs

- Shift (and Serial-Load)

- Combining Parallel-Load and Shift inputs

- Universal Register Design

# Combining Load and Clear Functions

Load function for a 1-bit register:

$$J = (\text{Input\_0}).(\text{Load})$$

$$K = (\text{Input\_0})'.(\text{Load})$$

Clear function for a 1-bit register:

$$J = 0$$

$$K = C$$

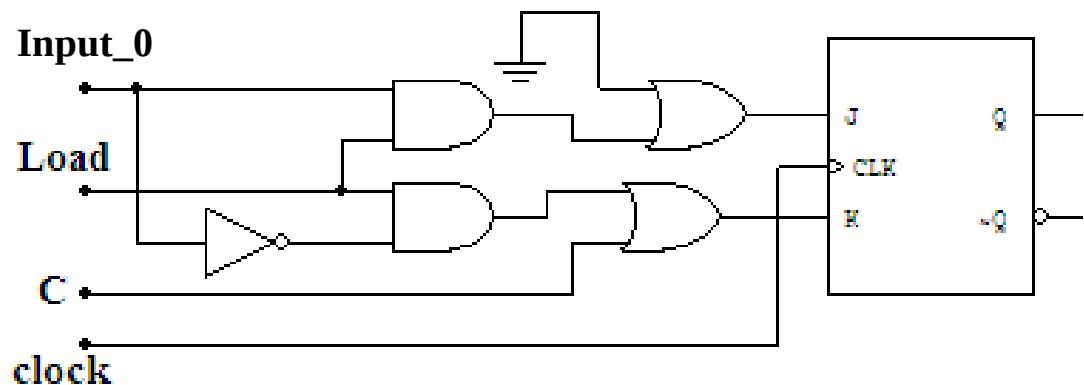
To combine these two functions, we use OR gates with 2 inputs.

To combine n different functions, we use OR gates with n inputs.

$$J = (\text{Input\_0}).(\text{Load}) + 0 = (\text{Input\_0}).(\text{Load})$$

$$K = (\text{Input\_0})'.(\text{Load}) + C$$

We assume that, C and L will never be “1” at the same time.



# Shift Right and Shift Left

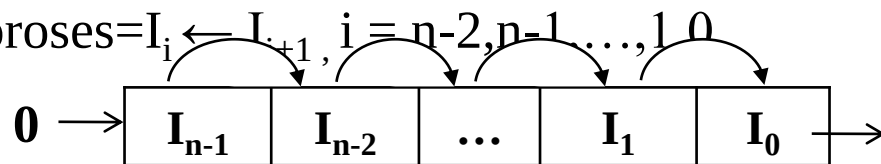
It is the process of shifting the contents of the register 1 bit to the left or right. As a result of shifting operations, there may be 3 different shifting operations depending on the value to be loaded into the least significant and most significant bits of the register.

## ▪ Shift by Zero

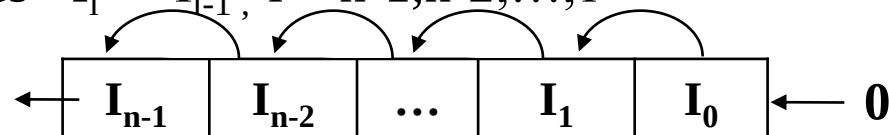
Depending on the direction of the shift, 0 is loaded into the LSB or MSB bit.

If the register content is  $(I_{n-1} I_{n-2} \dots I_1 I_0)$ ,

for shifting right proses= $I_i \leftarrow I_{i+1}, i = n-2, n-1, \dots, 1, 0$



for shifting left proses= $I_i \leftarrow I_{i-1}, i = n-1, n-2, \dots, 1$



## Shift with Zero Fill

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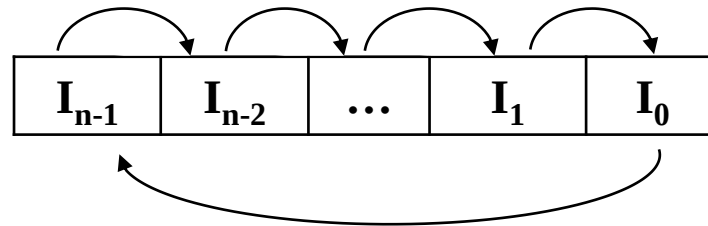
**Example:** Let's shift a 4-bit register to the right or left with zeros 4 times. Let the contents of the register be  $1101_2$ .

|             | 1101 |            | 1101 |
|-------------|------|------------|------|
| 1 bit right | 0110 | 1 bit left | 1010 |
| 1 bit right | 0011 | 1 bit left | 0100 |
| 1 bit right | 0001 | 1 bit left | 1000 |
| 1 bit right | 0000 | 1 bit left | 0000 |

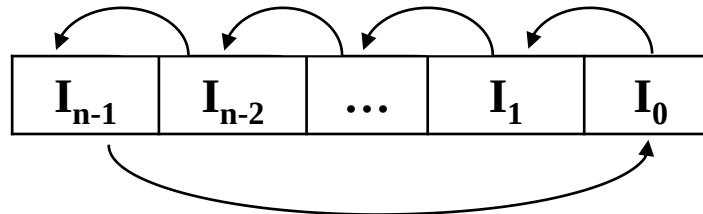
# Rotation

## ■ Rotation

In cyclic right shift, the LSB bit is transferred to the MSB bit. ( $I_{n-1} \leftarrow I_0$ ).



In cyclic left shift, the MSB bit is transferred to the LSB bit. ( $I_0 \leftarrow I_{n-1}$ ).



If an n-bit register is cyclically shifted n times, it will return to its initial state.

# Rotation

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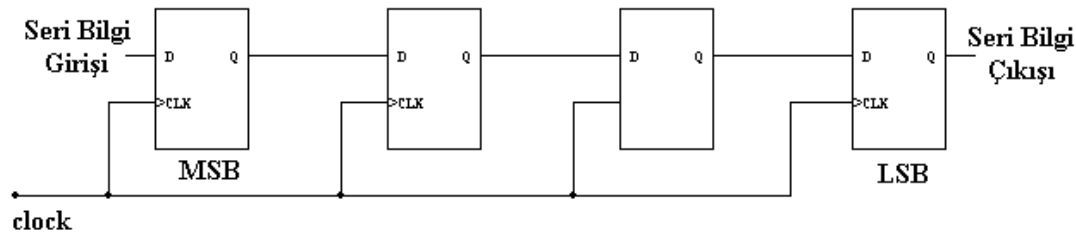
**Example:** Let's shift a 4-bit register cyclically left or right 4 times. Let the contents of the register be  $1101_2$ .

| 1101                    | 1101                    |
|-------------------------|-------------------------|
| 1 bit right <b>1110</b> | 1 bit left 101 <b>1</b> |
| 1 bit right <b>0111</b> | 1 bit left 01 <b>11</b> |
| 1 bit right <b>1011</b> | 1 bit left 1 <b>110</b> |
| 1 bit right <b>1101</b> | 1 bit left <b>1101</b>  |

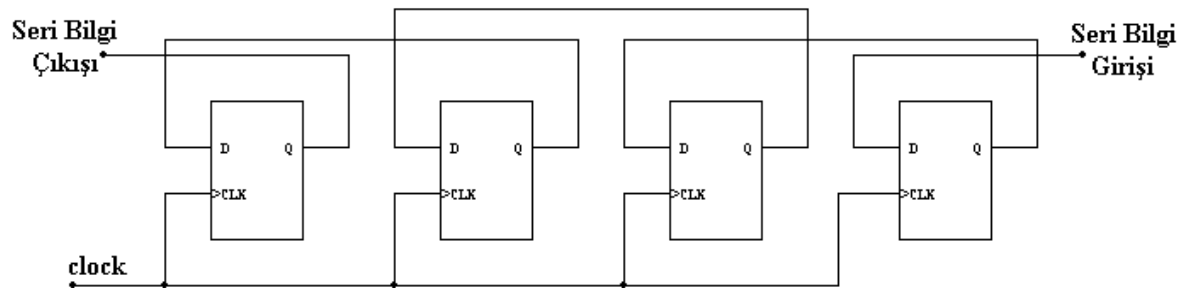
# Shift with Serial Input Fill

## ■ Shift with Serial Input/Info

- ⇒ When shifting to the right by entering serial information, a new value is loaded into the MSB bit and the old value of the LSB bit is lost.
- ⇒ When shifting left by entering serial information, a new value is loaded into the LSB bit and the old value of the MSB bit is lost.



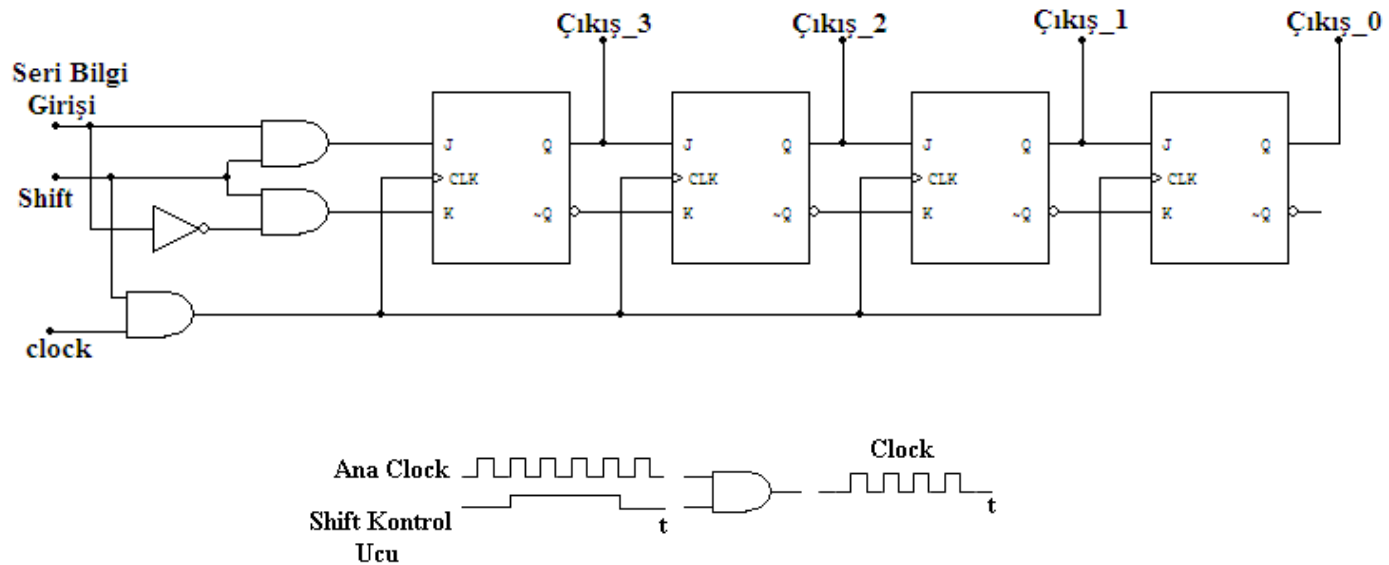
Shifting right



Shifting left

# Shift with Serial Input Fill

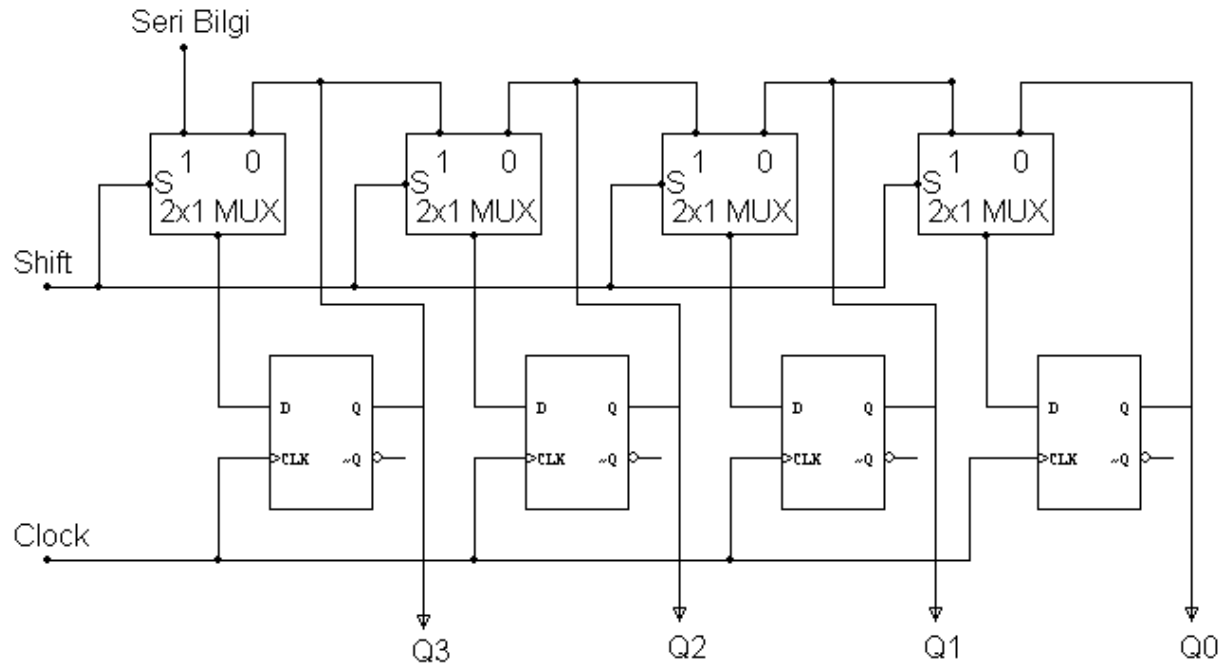
If we want to perform the shifting process in a controlled manner, we will need to add a Shift control pin to our circuit.



This circuit can also be used as serial input - serial output.

# Shift with Serial Input Fill

The following mechanism can be used to provide the right shift function with serial data input to the 4-bit register created from D-type flip flops;

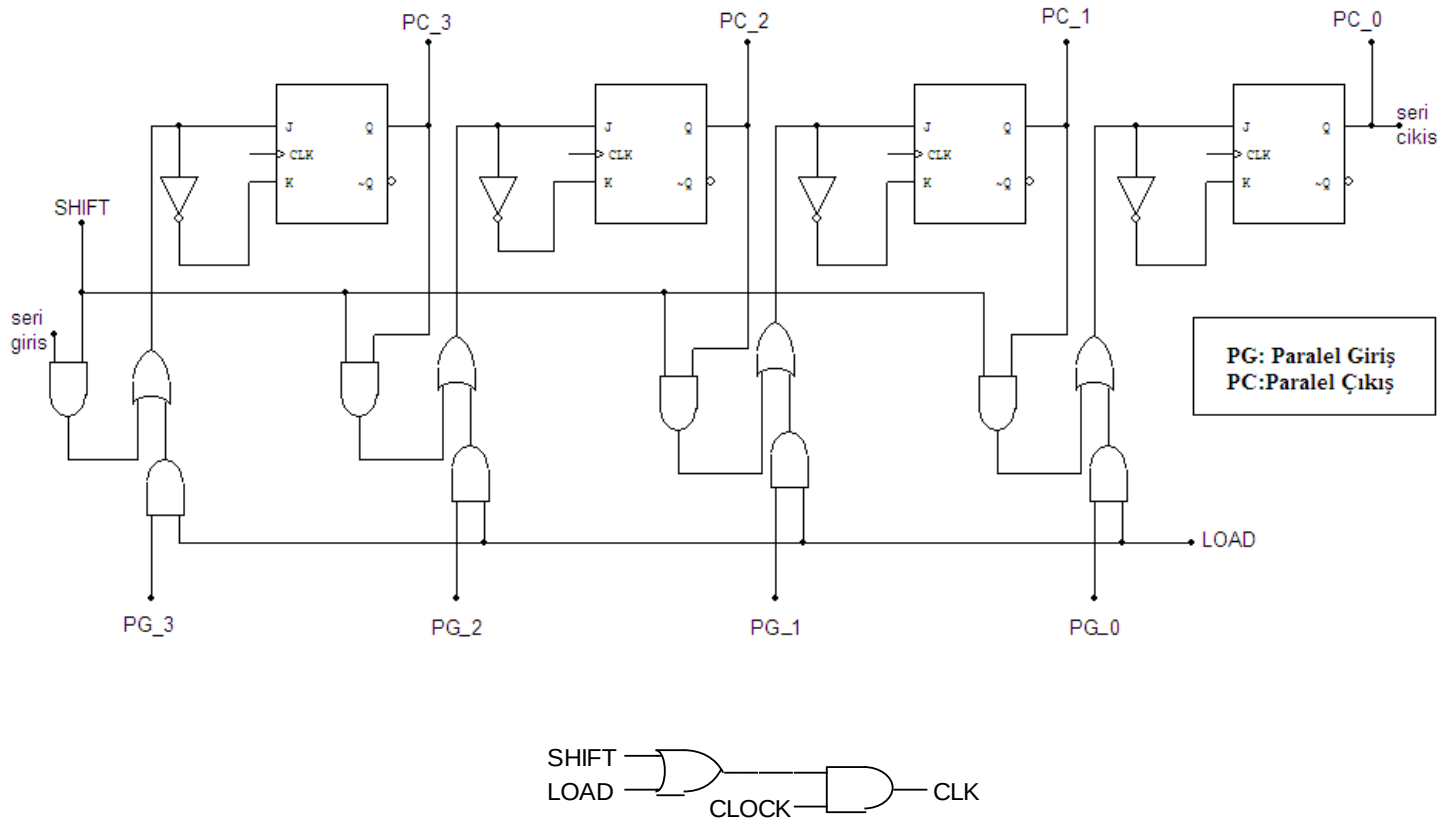


Shift = For 0 the register maintains its state,

Shift = For 1  $Q3 \leftarrow \text{Seri Input}$ ,  $Q2 \leftarrow Q3$ ,  $Q1 \leftarrow Q2$ ,  $Q0 \leftarrow Q1$

# Combining Parallel Load and Serial Load

In the following mechanism, the clock signal (CLK) is disabled to maintain the state of the recorder (both control inputs are not active).



## Combining Parallel Load and Serial Load

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Another way to combine these functions is to think of the shifting function as the load function. In this case, the inputs of the flip flops that make up the register are as follows;

$$J3 = (\text{LOAD}).(\text{PG\_3}) + (\text{SHIFT}).(\text{seri\_input\_data})$$

$$K3 = (\text{LOAD}).(\text{PG\_3})' + (\text{SHIFT}).(\text{seri\_input\_data})'$$

$$J2 = (\text{LOAD}).(\text{PG\_2}) + (\text{SHIFT}).(\text{PC\_3})$$

$$K2 = (\text{LOAD}).(\text{PG\_2})' + (\text{SHIFT}).(\text{PC\_3})'$$

$$J1 = (\text{LOAD}).(\text{PG\_1}) + (\text{SHIFT}).(\text{PC\_2})$$

$$K1 = (\text{LOAD}).(\text{PG\_1})' + (\text{SHIFT}).(\text{PC\_2})'$$

$$J0 = (\text{LOAD}).(\text{PG\_0}) + (\text{SHIFT}).(\text{PC\_1})$$

$$K0 = (\text{LOAD}).(\text{PG\_0})' + (\text{SHIFT}).(\text{PC\_1})'$$

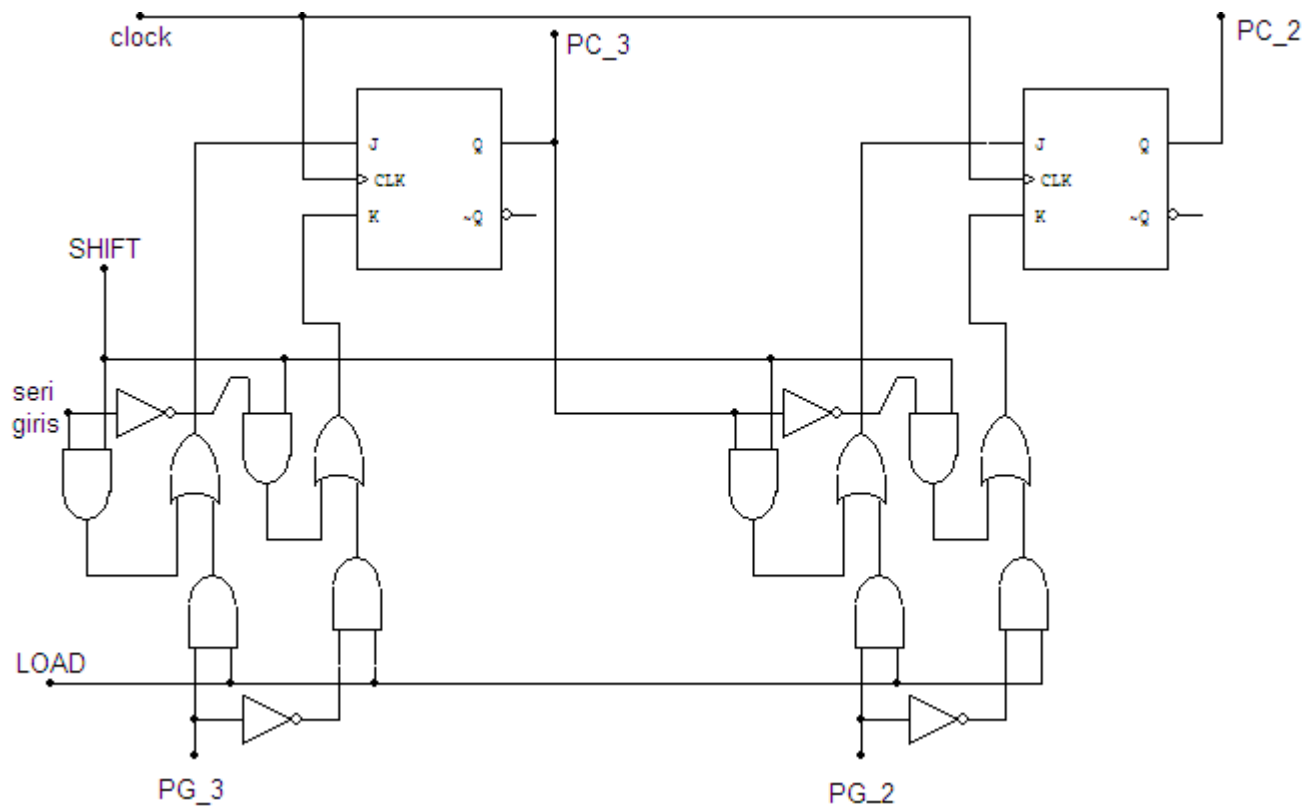
# Combining Parallel Load and Serial Load

$$J3 = (\text{LOAD}).(\text{PG}_3) + (\text{SHIFT}).(\text{seri input})$$

$$K3 = (\text{LOAD}).(\text{PG}_3)' + (\text{SHIFT}).(\text{seri input})'$$

$$J2 = (\text{LOAD}).(\text{PG}_2) + (\text{SHIFT}).(\text{PC}_3)$$

$$K2 = (\text{LOAD}).(\text{PG}_2)' + (\text{SHIFT}).(\text{PC}_3)'$$



Combining parallel loading and right shifting controls for bit 2 of the register

# Universal Register

| Choise      | Function                               |
|-------------|----------------------------------------|
| $S1S0 = 00$ | Normal storage status                  |
| $S1S0 = 01$ | Shifting right with serial information |
| $S1S0 = 10$ | Shifting left with serial information  |
| $S1S0 = 11$ | Loading data parallel                  |

